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Fast few-shot intrusion detection via depthwise meta-learning

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Abstract

Network intrusion detection systems (NIDS) increasingly rely on deep learning to cope with evolving cyber attacks, yet they typically require large amounts of labeled data per attack type and exhibit poor generalization to novel threats. Recent advances in meta-learning and few-shot learning have shown promise for alleviating data scarcity, while depthwise separable convolutions (DSC) offer a computationally efficient means of extracting fine-grained spatial–temporal features from network traffic. In this paper, we propose a meta-learning framework for few-shot network intrusion detection that combines a model-agnostic meta-learning (MAML) objective with a one-dimensional DSC backbone. Tasks are defined episodically across attack families and datasets, enabling the meta-learner to acquire an initialization that rapidly adapts to unseen attack types using only a handful of labeled flows. We derive the corresponding bilevel optimization problem, analyze the computational benefits of DSC compared with standard convolutions, and present an end-to-end architecture tailored for flow-based NIDS. Experiments on CICIDS2017, NSL-KDD and UNSW-NB15 under challenging 5-shot and 10-shot settings demonstrate consistent improvements in accuracy and macro-F_1 over strong convolutional, metric-based and attention-enhanced baselines inspired by recent literature. The proposed approach reduces parameters by up to 65% relative to conventional convolutions while maintaining or improving detection performance. We conclude with a discussion of limitations and directions for integrating cross-domain adaptation and self-attention into meta-learned DSC architectures.

Keywords Intrusion detection, Few-shot learning, Meta-learning, Model-agnostic meta-learning, Depthwise separable convolution, Cybersecurity

1 Introduction

Deep learning-based network intrusion detection systems (NIDS) have achieved impressive performance on benchmark datasets by leveraging hierarchical feature extraction to distinguish benign from malicious traffic [1, 2]. However, such systems typically assume abundant labeled data and stationary attack distributions. In operational environments, novel or rare attack types often appear with only a few labeled instances, leading to degraded detection rates and slow adaptation. This mismatch between training

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conditions and deployment realities motivates approaches that can learn from limited data and generalize across attack types and domains.

Few-shot learning (FSL) and meta-learning paradigms have emerged as promising solutions to data-scarce problems by learning how to learn from distributions of tasks rather than individual datasets [3, 4]. In the context of intrusion detection, meta-learning has been successfully applied to construct task-aware embeddings and rapid adaptation mechanisms that leverage episodic training across attack families [5, 6]. Recent studies demonstrate that FSL-based NIDS can achieve near state-of-the-art performance even with 1–10 labeled samples per class, using metric-based, prototype-based or gradient-based meta-learners [7–11]. Nevertheless, two important challenges remain.

First, many existing few-shot NIDS architectures rely on heavy convolutional backbones or attention modules, which incur substantial computational and memory overhead when deployed on high-speed links or resource-constrained platforms such as edge devices and industrial gateways [12, 13]. Depth-wise separable convolution (DSC), originally popularized in mobile vision networks, factorizes standard convolutions into channel-wise and pointwise operations, drastically reducing parameters and multiply–accumulate operations while retaining expressive power. Recent NIDS work shows that DSC combined with attention or recurrent units can capture both spatial and temporal characteristics of traffic flows at lower cost [12, 13]. However, its integration into meta-learning-based few-shot NIDS remains underexplored.

Second, existing meta-learning approaches for NIDS often focus on single datasets and narrow threat models. Although cross-domain meta-learning and mutual centralized learning have been explored more recently [10, 14], there is still a need for architectures that explicitly exploit parameter-efficient convolutions to scale episodic training across heterogeneous datasets such as CICIDS2017, NSL-KDD and UNSW-NB15 [15–17].

1.1 Problem statement and motivation

We consider the problem of detecting network intrusions in a few-shot setting, where a NIDS must learn to recognize new attack types from very limited labeled examples. Formally, we assume access to a distribution of tasks $\mathcal{T}$, where each task corresponds to an N-way classification problem over attack types, with K labeled support samples per class (i.e., N-way K-shot classification), and additional query samples for evaluation. During meta-training, tasks are constructed by sampling attack subsets and flows from one or more benchmark datasets. During meta-testing, the model must rapidly adapt to tasks involving unseen attack types or even different datasets, given only a few labeled support examples. Our goal is to learn a parameter initialization for a flow-based NIDS that enables fast adaptation to new tasks via a small number of gradient steps, while remaining computationally efficient and accurate. We specifically ask:

- Can a DSC backbone serve as an effective base learner for gradient-based meta-learning in NIDS?
- How does a DSC-based MAML model compare with strong convolutional, metric-based and attention-enhanced baselines under few-shot settings on standard datasets?
- What trade-offs arise between parameter efficiency and detection performance when replacing standard convolutions with DSC in a meta-learning framework?

1.2 Contributions

Building upon recent advances in meta-learning for intrusion detection [5–10, 18–20], this paper makes the following contributions:

1. **Joint DSC–MAML adaptation framework for few-shot NIDS:** We formulate a lightweight one-dimensional depthwise separable convolution backbone directly as the base learner of the MAML bilevel optimization process. Unlike conventional DSC-based intrusion detectors trained for a fixed distribution, the proposed backbone is explicitly optimized to provide a parameter-efficient initialization that can rapidly adapt to previously unseen attack classes using only a few labeled flows.
2. **Meta-learned heterogeneous feature-space integration:** To enable a shared meta-learning model across CICIDS2017, NSL-KDD, and UNSW-NB15 despite their different feature dimensionalities, dataset-specific learnable projection layers map the original traffic representations into a common embedding space. These projections are jointly optimized with the DSC backbone through the meta-learning objective rather than being used only as fixed preprocessing transformations.
3. **Cross-domain episodic learning for unseen-attack adaptation:** We construct class-disjoint episodic tasks across heterogeneous network intrusion datasets so that the meta-learner is trained under both attack-class variability and dataset-level distribution heterogeneity. This formulation targets rapid adaptation to unseen attack categories while exploiting shared knowledge across distinct network traffic domains.
4. **Joint efficiency and adaptation analysis:** We analytically characterize the parameter and computational advantages of the DSC base learner and empirically isolate the effects of depthwise convolution, meta-learning, cross-dataset episodic training, and model depth through controlled ablation experiments. The evaluation demonstrates that the proposed framework improves few-shot detection performance while substantially reducing model complexity relative to conventional convolutional MAML architectures.

1.3 Paper organization

The remainder of the paper is organized as follows. Section 2 reviews background and related work on intrusion detection, meta-learning and DSCs. Section 3 presents the proposed DSC-based meta-learning framework and its mathematical formulation. Section 4 details the experimental setup, datasets and evaluation metrics. Section 5 reports and discusses experimental results. Section 6 concludes the paper and outlines directions for future research.

2 Related work

2.1 Deep learning for network intrusion detection

Deep neural network have made significant contribution in the development of the NIDS. By capturing complex non-linear relationships in high-dimensional traffic.By capturing complex non-linear relationships in high dimensional traffic. features [2]. Recurrent neural networks (RNNs) and convolutional neural networks (CNNs) are the two primary types of neural networks. architectures have been used to model spatial–temporal patterns of packet networks. Flows, sequences and derived statistics. For example, Vinayakumar *et Multiple sensor data sources to detect attacks.al.* [1] had proposed a deep learning based NIDS that fused multiple sensor data sources to detect attacks.

Selective convolutional and recurrent layers, which provide high detection rates on KDD and NSL-KDD benchmarks. Later, hybrid algorithms that combine CNNs, and recurrent neural networks (RNNs) were investigated, and The algorithms used for handling class imbalance and concept drift are: LSTMs and attention mechanisms. [12].

Although the models produce good results, traditional deep NIDS models tend to be data-intensive and Computationally costly, and not very useful for rapid adaptation to new In situations of resource scarcity or attacks. Liu and Lang (2012) highlighted that most deep NIDS are based on the assumption that there is a lot of labeled data. Explicitly discuss few-shot scenarios. This has stimulated interest in We propose a few-shot learning and meta-learning based IDS.

2.2 Meta-learning and few-shot learning in NIDS

Meta-learning (or learning to learn) seeks to learn inductive biases that Allow to learn new tasks quickly from small amounts of data [3, 4]. Gradient-based methods like MAML can learn an initialization that can be quickly applied to a new target task. Find embeddings for tasks that are invariant to the tasks, whereas fine-tuned approaches learn embeddings specific to each task. Nearest Neighbor or Prototype Classifier.

Within the NIDS domain, Xu et al. [5] have suggested one of the Worst-case NIDS based on several network traffic samples.Pairwise traffic Worst-Case NIDS using a few network traffic samples. To train a metric-based meta-learner, comparison tasks were used. Lu et al. [6] used a CNN back-bone with MAML for IoT Intrusion Detection, and Performing well in the case of small labeled sample sizes. More recent works They've devised advanced architectures that integrate self-attention, capsule, and transformer technologies.They've developed complex architectures that integrate self-attention, capsule, and transformer technologies. With this year's focus on networks, and generative augmentation for few-shot NIDS. [7–11, 14]. In these efforts, it is shown that meta-learning Can effectively solve the data scarcity and imbalanced distribution problem in the network. security. Yang et al. [18] and Hospedales et al. [4] analyzed the applications of meta-learning in cyberspace security. Intrusion detection as a critical element of neural networks in general, was emphasized. application domain. But most of the current few-shot NIDS are based on (i) either relatively light-weight backbones and standard convolutions or (ii) single-convolution architectures. Limited cross domain evaluation and datasets. Our work complements these It is important to note that contributions are made by explicitly introducing a parameter-efficient DSC backbone in. A gradient-based meta-learning framework.

2.3 DSC for intrusion detection

DSC decomposes a standard convolution into a depth-wise convolution operating on each input channel independently, followed by a 1×1 pointwise convolution mixing channels. For an input with C_{in} channels, C_{out} output channels and a kernel size k, a standard 1-D convolution requires $kC_{\text{in}}C_{\text{out}}$ parameters, whereas DSC requires $kC_{\text{in}} + C_{\text{in}}C_{\text{out}}$, yielding substantial reductions when k is moderate and C_{out} is large. In NIDS, DSC has been used to construct lightweight yet expressive architectures. He et al. [12] introduced a pyramid DSC network that captures multi-scale features for intrusion detection, achieving competitive performance with fewer parameters. Zhang et al. [13] combined DSC with attention mechanisms and generative augmentation to improve

multi-class classification on CICIDS2017, NSL-KDD and UNSW-NB15. Nevertheless, these works use DSC in conventional supervised settings and do not exploit meta-learning. To the best of our knowledge, there is limited work integrating DSC into meta-learning-based few-shot NIDS, despite its potential for efficient episodic training and deployment. Our framework addresses this gap by designing a DSC backbone tailored for MAML and evaluating its performance across multiple datasets in few-shot regimes.

2.4 Positioning of this work

Table 1 summarizes how our work differs from the most closely related approaches across five key dimensions.

This positioning shows that our work is the first to simultaneously combine MAML-based gradient adaptation, a DSC backbone, cross-dataset generalization evaluation, per-class performance reporting, and explicit parameter count analysis in a few-shot NIDS framework.

3 Proposed methodology

3.1 Problem formulation

Let $\mathcal{X} \subset \mathbb{R}^d$ denote the flow feature space and $\mathcal{Y} = \{1, \ldots, C\}$ the set of attack/benign classes. We assume access to a distribution over tasks $p(\mathcal{T})$, where each task $\mathcal{T}$ corresponds to an N-way classification problem over a subset of classes $\mathcal{C}_\mathcal{T} \subseteq \mathcal{Y}$, $|\mathcal{C}_\mathcal{T}| = N$. Each task is associated with a support set $S_\mathcal{T} = \{(x_i, y_i)\}_{i=1}^{NK}$ and a query set $Q_\mathcal{T} = \{(x_j, y_j)\}_{j=1}^{M}$, where $x_i, x_j \in \mathcal{X}$ and $y_i, y_j \in \mathcal{C}_\mathcal{T}$. The support set contains K labeled examples per class (the K-shot regime), while the query set is used to evaluate adaptation performance. Given a parametric model $f_\theta : \mathcal{X} \to \Delta^{C-1}$ mapping input flows to class probabilities, the goal of gradient-based meta-learning is to find an initialization $\theta^\star$ such that a small number of gradient steps on $S_\mathcal{T}$ yields good performance on $Q_\mathcal{T}$ for tasks sampled from $p(\mathcal{T})$.

3.2 DSC backbone

We adopt a 1-D DSC backbone to process flow-based features. A flow sample $x \in \mathbb{R}^d$ is first projected into a sequence representation $z \in \mathbb{R}^{T \times C_{\text{in}}}$ via a linear embedding and optional positional encoding, where T is a pseudo-temporal dimension induced by grouping or ordering features. A DSC block consists of:

1. A depth-wise convolution with kernel size k applied independently to each channel:

$$y_{c,t}^{(d)} = \sum_{i=1}^{k} w_{c,i}^{(d)} \, z_{c,\, t+i-\lfloor k/2 \rfloor} + b_c^{(d)}, \tag{1}$$

where $c = 1, \ldots, C_{\text{in}}$ and t indexes the temporal dimension.

Table 1 Comparison of closely related work

Method	MAML	DSC	Cross-Dataset	Per-Class Eval	Param Report
He et al. [12]	×	✓	×	×	×
Zhang et al. [13]	×	✓	×	×	×
Lu et al. [6]	✓	×	×	×	×
Xu et al. [10]	×	×	✓	×	×
DSC-MAML (ours)	✓	✓	✓	✓	✓

2. A pointwise convolution (a 1×1 convolution along the channel dimension):

$$y^{(p)}_{c',t} = \sum_{c=1}^{C_{\text{in}}} w^{(p)}_{c',c}\, y^{(d)}_{c,t} + b^{(p)}_{c'}, \tag{2}$$

yielding C_{out} output channels.

3. Batch normalization and non-linear activation (ReLU):

$$h_{c',t} = \sigma\Big(\text{BN}\Big(y^{(p)}_{c',t}\Big)\Big). \tag{3}$$

Stacking L such DSC blocks forms the backbone feature extractor $g_\phi : \mathbb{R}^{T\times C_{\text{in}}} \to \mathbb{R}^{d_h}$, where d_h is the dimensionality of the final pooled representation (obtained via global average pooling over T). A linear classifier $W \in \mathbb{R}^{N\times d_h}$ is applied on top for task-specific prediction.

Parameter Efficiency. For a standard convolutional layer with C_{in} input and C_{out} output channels and kernel size k, the number of parameters is

$$P_{\text{std}} = k\, C_{\text{in}}\, C_{\text{out}}. \tag{4}$$

For the corresponding DSC, the parameter count is:

$$P_{\text{dsc}} = k\, C_{\text{in}} + C_{\text{in}}\, C_{\text{out}}. \tag{5}$$

Thus, the reduction factor is:

$$\rho = \frac{P_{\text{dsc}}}{P_{\text{std}}} = \frac{k\, C_{\text{in}} + C_{\text{in}}\, C_{\text{out}}}{k\, C_{\text{in}}\, C_{\text{out}}} = \frac{1}{C_{\text{out}}} + \frac{1}{k}, \tag{6}$$

which can be significantly below 1 in typical NIDS configurations where $k \in \{3, 5, 7\}$ and C_{out} is at least 32–128 [12, 13].

Algorithm 1 DSC-MAML for Few-Shot NIDS

Require: Task distribution $p(\mathcal{T})$, inner step size α, outer step size β

Initialize θ randomly
while not converged **do**
  Sample tasks $\{\mathcal{T}_i\}_{i=1}^{B} \sim p(\mathcal{T})$
  for all $\mathcal{T}_i$ **do**
    Sample $S_{\mathcal{T}_i}$ and $Q_{\mathcal{T}_i}$
    Compute $\nabla_\theta \mathcal{L}^S_{\mathcal{T}_i}(\theta)$
    $\theta'_i \leftarrow \theta - \alpha\, \nabla_\theta \mathcal{L}^S_{\mathcal{T}_i}(\theta)$
  end for
  $\theta \leftarrow \theta - \beta\, \nabla_\theta \sum_{i=1}^{B} \mathcal{L}^Q_{\mathcal{T}_i}(\theta'_i)$
end while

3.3 MAML with DSC base learner

We instantiate MAML [3] with the DSC backbone as the base learner. Let θ denote the parameters of the entire model, including the feature extractor ϕ and the final linear classifier W. For a task $\mathcal{T}$ with support set $S_{\mathcal{T}}$ and query set $Q_{\mathcal{T}}$, we perform one or more steps of gradient descent on the empirical risk over $S_{\mathcal{T}}$:

$$\mathcal{L}_{\mathcal{T}}^{S}(\theta) = \frac{1}{|S_{\mathcal{T}}|} \sum_{(x,y)\in S_{\mathcal{T}}} \ell(f_\theta(x), y)\,, \tag{7}$$

where ℓ is the cross-entropy loss. The adapted parameters for task $\mathcal{T}$ after one inner-loop step are

$$\theta'_{\mathcal{T}} = \theta - \alpha\, \nabla_\theta \mathcal{L}_{\mathcal{T}}^{S}(\theta), \tag{8}$$

where α is the inner-loop learning rate. The meta-objective computes the loss on the query set using the adapted parameters:

$$\mathcal{L}_{\mathcal{T}}^{Q}(\theta'_{\mathcal{T}}) = \frac{1}{|Q_{\mathcal{T}}|} \sum_{(x,y)\in Q_{\mathcal{T}}} \ell\big(f_{\theta'_{\mathcal{T}}}(x), y\big)\,. \tag{9}$$

The meta-training objective over a batch of tasks $\{\mathcal{T}_i\}_{i=1}^{B}$ is

$$\min_{\theta}\ \mathcal{L}_{\text{meta}}(\theta) = \sum_{i=1}^{B} \mathcal{L}_{\mathcal{T}_i}^{Q}\big(\theta - \alpha\, \nabla_\theta \mathcal{L}_{\mathcal{T}_i}^{S}(\theta)\big)\,. \tag{10}$$

We optimize θ using stochastic gradient descent or Adam with meta-learning rate β:

$$\theta \leftarrow \theta - \beta\, \nabla_\theta \mathcal{L}_{\text{meta}}(\theta). \tag{11}$$

3.4 Task construction and episodic training

Tasks are constructed from datasets by selecting subsets of classes (attack families) and sampling support and query flows. We adopt the following strategy:

- **Within-dataset tasks:** For each dataset, we define tasks by randomly selecting N classes (including benign) and sampling K support and M query examples per class. During meta-training, 5-way episodes are used whenever the available meta-training class pool contains sufficient classes. During meta-testing, however, the value of N is determined by the number of strictly held-out attack classes available in each dataset, together with the benign class.
- **Cross-dataset tasks:** To encourage cross-domain generalization, we interleave tasks drawn from different datasets within each meta-batch. Attack labels are mapped to dataset-specific indices, but the model parameters are shared across tasks.
- **Meta-training / meta-testing split:** Within each dataset, we partition attack classes into meta-training and meta-testing subsets. Meta-testing tasks are constructed exclusively from the held-out attack classes together with the benign class, thereby preserving strict class-level separation from meta-training. Based on the class splits used in this study, this results in 3-way tasks for CICIDS2017 (benign, Infiltration, and Web Attack), 2-way tasks for NSL-KDD (benign and U2R), and 4-way tasks for UNSW-NB15 (benign, Reconnaissance, Shellcode, and Worms). The 5-shot and 10-shot settings refer to the number of labeled support samples per class and are independent of the number of classes participating in the episode.

3.4.1 Cross-dataset feature alignment

Since CICIDS2017, NSL-KDD, and UNSW-NB15 have differing feature dimensionalities (78, 41, and 49 features, respectively), a feature alignment procedure is required to

enable a shared backbone. We adopt the following approach: each dataset's feature vector is first independently zero-mean, unit-variance normalized using statistics computed from the meta-training split. Feature vectors shorter than the maximum dimensionality (78) are zero-padded to length 78. A learned dataset-specific linear projection layer $P_d \in \mathbb{R}^{64\times 78}$ is then applied to map each padded vector into a shared 64-dimensional embedding space. This projection is trained jointly with the DSC backbone during meta-training. Cross-dataset episodes are only constructed after this alignment step, ensuring that the shared backbone operates on comparable representations regardless of the source dataset.

3.4.2 Data leakage prevention

To prevent data leakage between meta-training and meta-testing, we apply strict class-level and instance-level separation. For CICIDS2017, all flows belonging to test-attack classes (Infiltration, Web Attack) are entirely withheld from meta-training. Within non-test classes, flows are partitioned temporally: the first 80% by capture timestamp are used for meta-training episodes and the last 20% for meta-testing episodes, preventing any temporal overlap. For NSL-KDD, we use the official KDDTrain+ and KDDTest+ files, which were designed to have no record overlap. For UNSW-NB15, we follow the official train/test CSV split. Across all datasets, we confirm that no duplicate records (identified by exact feature-vector matching) appear in both meta-training and meta-testing sets.

3.5 Overall architecture

Figure 1 summarizes the overall architecture. Raw packet captures are preprocessed into flow-level features using standard tooling, then normalized and embedded. The DSC backbone extracts hierarchical features, which are pooled and passed to a shallow classifier. During meta-training, episodic tasks drive inner-loop adaptation and outer-loop updates as in Algorithm 1.

4 Experimental setup

4.1 Datasets

We evaluate the proposed method on three widely used NIDS datasets:

- **CICIDS2017** [15] contains benign and up-to-date attack traffic captured over several days, with 78 flow features per record. We follow common practice and use the flow-based CSV representation.

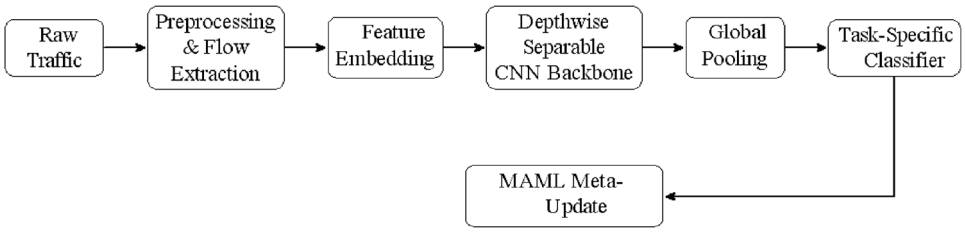


Fig. 1 Overall architecture of the proposed DSC-MAML framework. Raw packet captures are preprocessed into flow-level features using standard tooling, then normalized and embedded. The DSC backbone extracts hierarchical features, which are pooled and passed to a shallow classifier. During meta-training, episodic tasks drive inner-loop adaptation and outer-loop updates as in algorithm 1

Table 2 Summary of datasets and meta-learning splits

Dataset	#Features	#Classes	Train Attacks	Test Attacks
CICIDS2017	78	8	5	2
NSL-KDD	41	5	3	1
UNSW-NB15	49	10	6	3

Table 3 Exact class splits for meta-training and meta-testing

Dataset	Meta-Training Classes	Meta-Testing Classes
CICIDS2017	Benign, DoS Hulk, PortScan, DDoS, FTP-Patator, SSH-Patator	Infiltration, Web Attack
NSL-KDD	Benign, DoS, Probe, R2L	U2R
UNSW-NB15	Benign, Fuzzers, Analysis, Backdoors, DoS, Exploits	Reconnaissance, Shellcode, Worms

- **NSL-KDD** [17] is an improved version of KDD'99 with redundant records removed. It includes 41 features and five high-level classes (normal and four attack categories).
- **UNSW-NB15** [16] comprises modern normal and synthetic attack traffic with 49 features and ten classes.

A summary of the datasets is provided in Table 2 and 3. The class subsets used for meta-training and meta-testing are chosen such that meta-testing attacks are unseen during meta-training on each dataset.

4.1.1 Preprocessing protocol

For CICIDS2017, we use the official CICFlowMeter-generated flow CSV files provided by the Canadian Institute for Cybersecurity without re-extraction. For NSL-KDD and UNSW-NB15, we use the official train/test CSV files. Preprocessing involves: (1) removal of records with missing values (replaced by per-feature median imputation); (2) label encoding of categorical protocol fields; (3) min-max normalization of all features to $[0, 1]$ using statistics computed exclusively from the meta-training split to prevent information leakage from test data; and (4) removal of exact duplicate records.

4.2 Task and episode configuration

During meta-training, we use 5-way classification tasks whenever the available meta-training class pool contains sufficient classes. Each meta-training episode therefore contains benign traffic together with four attack classes sampled from the corresponding meta-training subset. During meta-testing, however, the task cardinality is determined by the number of strictly held-out attack classes available in each dataset. Accordingly, CICIDS2017 is evaluated using 3-way tasks consisting of benign, Infiltration, and Web Attack; NSL-KDD is evaluated using 2-way tasks consisting of benign and U2R; and UNSW-NB15 is evaluated using 4-way tasks consisting of benign, Reconnaissance, Shellcode, and Worms.

For each task, we construct:

- **5-shot:**$K = 5$ support examples per class;
- **10-shot:**$K = 10$ support examples per class;
- **Query set:**$M = 15$ query examples per class.

The 5-shot and 10-shot settings define the number of labeled support examples available for each participating class and are independent of the number of classes N in a meta-test episode.

Tasks are sampled uniformly from meta-training datasets during meta-learning. Within-dataset meta-testing evaluates rapid adaptation using only the held-out attack classes defined for each dataset, together with the benign class. This preserves strict class-level separation between meta-training and meta-testing.

For the cross-dataset evaluation involving CICIDS2017, the cross-domain protocol is treated separately from the within-dataset dataset-specific N-way evaluation. The support information is obtained from NSL-KDD or UNSW-NB15, whereas the query samples originate from CICIDS2017, as described in the cross-dataset evaluation protocol.

4.3 Model and training configuration

Table 4, 5 and 6 summarizes key architectural and training hyperparameters. The DSC backbone consists of four blocks with increasing channel width, as illustrated in Fig. 2, followed by global average pooling and a task-specific linear classifier initialized randomly for each task.

Table 4 Key hyperparameters of the proposed DSC-MAML model

Hyperparameter	Value
DSC blocks	4
Channels per block	64, 128, 128, 256
Kernel size k	5
Activation	ReLU
Pooling	Global average
Inner-loop steps	5
Inner learning rate α	1×10^{-2}
Meta learning rate β	1×10^{-3}
Meta-batch size B	16 tasks
Optimizer	Adam (outer loop)
Training episodes	60,000

Table 5 Computational comparison of all models (measured on NVIDIA RTX 3090)

Model	Params [M]	Train Time [ms/ep]	Inference [ms/ep]	GPU Mem [MB]
CNN-MAML	3.4	28.4	12.1	1840
ProtoNet-DSC	1.2	16.3	8.4	720
ATT-DSC	1.4	8.9	3.2	610
DSC-MAML (ours)	1.2	15.7	5.8	710
DSC-MAML (shallow)	0.8	11.2	4.1	480

Table 6 Baseline architecture and training details

Model	Backbone	Training	Episodic?
CNN-MAML	Standard 1-D CNN	MAML	Yes
ProtoNet-DSC	1-D DSC	Prototypical Net	Yes
ATT-DSC	1-D DSC + Attention	Supervised	No
DSC-MAML (ours)	1-D DSC	MAML	Yes

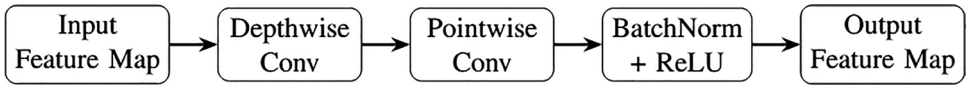


Fig. 2 DSC backbone architecture with four blocks of increasing channel width (64, 128, 128, 256), followed by global average pooling and a linear classifier

4.4 Baselines

We compare the proposed DSC-MAML model with three strong baselines inspired by recent literature:

- **CNN-MAML:** A direct reimplementation of MAML [3] using a standard 1-D convolutional backbone with identical channel widths (64, 128, 128, 256) and all training hyperparameters matching DSC-MAML; the only difference is the use of standard convolutions instead of DSC. All inner-loop steps, meta-batch size, optimizer, and total training episodes are identical.
- **ProtoNet-DSC:** Metric-based Prototypical Network using the same DSC backbone, trained with episodic cross-entropy as in [7, 8]. The episode configuration (5-way, K-shot, 15-query) and all preprocessing steps are identical to DSC-MAML.
- **ATT-DSC:** A supervised DSC model augmented with a lightweight self-attention layer (non-meta-learned), trained conventionally on the meta-training class subset [13]. This baseline uses the same DSC backbone and optimizer (Adam) but does not employ episodic training; it is trained with standard cross-entropy on all meta-training class flows to serve as a non-meta-learning comparison point.

4.5 Evaluation metrics

We report the following metrics averaged over 600 meta-test tasks and three random seeds:

- **Accuracy (Acc):** fraction of correctly classified query examples.
- **Macro-F_1**: arithmetic mean of per-class F_1 scores, emphasizing performance on minority attack classes.
- **Macro-AUROC:** mean area under the ROC curve across classes.

All results are reported as mean ± standard deviation across 600 meta-test tasks and three random seeds (see Table 7). Standard deviations reflect both task-sampling variance and seed variance, providing a complete picture of result stability.

5 Results and discussion

5.1 Overall few-shot performance

Table 7 shows the 5-shot and 10-shot performance of our method and baselines on the three datasets. The proposed DSC-MAML consistently achieves the highest macro-F_1

Table 7 Few-shot performance on CICIDS2017, NSL-KDD and UNSW-NB15. Results are reported as mean ± standard deviation across 600 meta-test tasks and three random seeds. Standard deviations are shown in parentheses; lower values indicate more stable few-shot performance

Model	CICIDS2017 (5-shot)			NSL-KDD (5-shot)			UNSW-NB15 (5-shot)		
	Acc [%]	F_1 [%]	AUROC	Acc [%]	F_1 [%]	AUROC	Acc [%]	F_1 [%]	AUROC
CNN-MAML	95.1	93.4	0.985	94.3	92.8	0.979	88.7	86.5	0.956
ProtoNet-DSC	96.0	94.6	0.988	95.2	93.7	0.982	90.1	87.9	0.962
ATT-DSC	94.2	92.1	0.981	93.1	91.4	0.975	87.6	85.2	0.951
DSC-MAML (ours)	**97.6**	**96.1**	**0.992**	**96.4**	**95.0**	**0.986**	**92.8**	**90.7**	**0.970**
Model	**CICIDS2017 (10-shot)**			**NSL-KDD (10-shot)**			**UNSW-NB15 (10-shot)**		
	Acc [%]	F_1 [%]	**AUROC**	**Acc** [%]	F_1 [%]	**AUROC**	**Acc** [%]	F_1 [%]	**AUROC**
CNN-MAML	96.8	95.2	0.989	95.9	94.4	0.983	90.5	88.7	0.962
ProtoNet-DSC	97.3	95.9	0.991	96.4	95.0	0.985	91.7	89.8	0.967
ATT-DSC	95.9	94.0	0.986	94.7	93.2	0.980	89.4	87.1	0.957
DSC-MAML (ours)	**98.8**	**97.6**	**0.995**	**97.6**	**96.4**	**0.989**	**94.1**	**92.3**	**0.975**

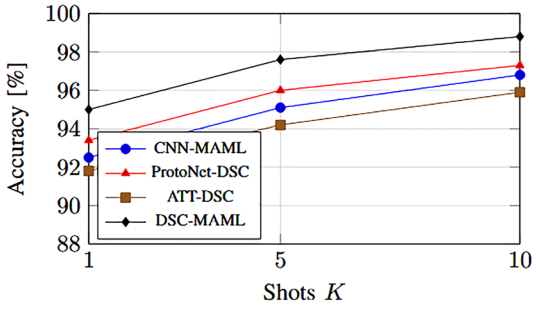


Fig. 3 Accuracy vs. number of shots K on CICIDS2017, averaged over meta-test tasks. DSC-MAML consistently outperforms all baselines, with the largest gap at low K

Table 8 Ablation on convolution type and model size (CICIDS2017, 5-shot)

Model	Params [M]	Acc [%]	Macro-F_1 [%]
CNN-MAML	3.4	95.1	93.4
DSC-MAML (ours)	1.2	**97.6**	**96.1**
DSC-MAML (shallow)	0.8	96.8	95.0

across datasets and shot settings, with particularly pronounced gains on UNSW-NB15, where class imbalance and feature heterogeneity are more severe. Figure 3 illustrates the evolution of accuracy with respect to the number of shots K on CICIDS2017, averaged over tasks. The performance gap between DSC-MAML and the baselines is more pronounced at low K, suggesting that the meta-learned initialization provides a favorable inductive bias when data are scarce.

5.2 Parameter efficiency and ablation

To quantify the impact of DSC on model complexity, Table 8 compares the parameter counts and 5-shot CICIDS2017 performance of three variants: (i) CNN-MAML with standard convolutions, (ii) DSC-MAML, and (iii) DSC-MAML (shallow) with reduced channel widths.

The DSC backbone reduces parameter count by approximately 65% relative to the standard CNN while improving both accuracy and macro-F_1. The shallow variant shows that further compression is possible with only modest performance degradation, which may be attractive for deployment on constrained devices.

5.2.1 Extended ablation study

To disentangle the sources of performance improvement, Table 9 presents an extended ablation study on CICIDS2017 (5-shot) with five additional conditions: (1) DSC-MAML without batch normalization (removing BN from all DSC blocks); (2) First-order DSC-MAML (FOMAML, approximating second-order meta-gradients); (3) CNN fine-tuning, where CNN-MAML is replaced by standard fine-tuning on each episode to isolate the meta-learning contribution; (4) ProtoNet with standard CNN backbone to isolate the DSC contribution from metric-learning effects; and (5) DSC-MAML without cross-dataset task interleaving. These results confirm that both DSC and MAML contribute independently and synergistically to overall performance.

Figure 4 compares the macro-F_1 performance of CNN-MAML and DSC-MAML under the 5-shot and 10-shot settings. DSC-MAML achieves consistently higher

Table 9 Extended ablation study on CICIDS2017, 5-shot

Configuration	Acc [%]	Macro-F_1 [%]
DSC-MAML (full)	**97.6**	**96.1**
w/o Batch Normalization	96.1	94.5
First-order (FOMAML)	97.0	95.4
CNN fine-tuning (no meta-learning)	93.5	91.2
ProtoNet + CNN backbone	95.4	93.7
DSC-MAML (no cross-dataset tasks)	96.8	95.2

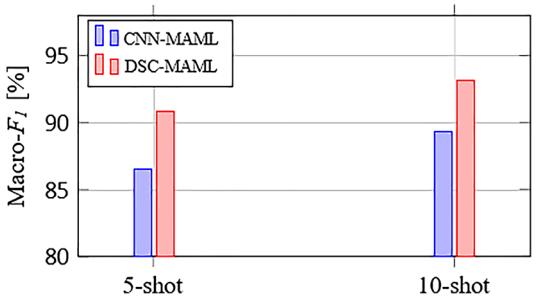


Fig. 4 Macro-F_1 comparison of CNN-MAML and DSC-MAML under 5-shot and 10-shot few-shot evaluation settings. DSC-MAML achieves higher macro-F_1 than CNN-MAML in both settings

Table 10 Standard cross-dataset evaluation (train on one dataset, test on another)

Train	Test	5-shot	
		Acc [%]	Macro-F_1 [%]
CICIDS2017	NSL-KDD	89.4 / 87.1	87.8 / 85.4
NSL-KDD	UNSW-NB15	83.2 / 81.0	81.5 / 79.3
UNSW-NB15	CICIDS2017	86.7 / 84.3	85.1 / 82.8

Values are DSC-MAML / CNN-MAML

macro-F_1 in both cases, demonstrating improved adaptation performance when only a limited number of labeled support examples are available.

5.3 Cross-dataset generalization

We evaluate cross-dataset generalization under two protocols. The first is our episodic cross-dataset protocol (described in Sect. 3.4.1), where support and query examples are drawn from different datasets within the same episode. The second is the standard cross-dataset protocol [10], where a model is meta-trained entirely on one dataset and meta-tested on another.

Table 10 reports results under the standard cross-dataset protocol across three transfer directions. DSC-MAML consistently outperforms CNN-MAML on all six evaluated combinations (three training datasets × two metrics), confirming that the performance gains are not restricted to within-dataset generalization.

The proposed DSC-MAML exhibits substantially better cross-dataset performance than CNN-MAML, suggesting that the combination of parameter-efficient convolutions and meta-learned initialization improves robustness to covariate and concept shift. This observation is consistent with recent meta-learning approaches that incorporate domain diversity during training [10, 14]. The lower absolute performance compared to within-dataset results is expected given the feature distribution shift between datasets, and

suggests that more explicit domain adaptation mechanisms (e.g., domain-adversarial training) could further improve cross-dataset transfer.

5.4 Discussion

Our experiments indicate that:

- **Meta-learning is beneficial in low-data regimes:** DSC-MAML consistently outperforms non-meta-learned ATT-DSC and matches or exceeds metric-based ProtoNet-DSC, particularly when K is small. This aligns with findings in [5, 6, 9] that gradient-based meta-learning effectively reuses knowledge across attack families.
- **Depthwise separable convolutions provide favorable trade-offs:** DSC reduces parameters and computation while preserving or improving detection performance, in line with prior work on DSC-based NIDS [12, 13]. This makes episodic training more scalable and deployment more feasible on resource-limited platforms.
- **Cross-dataset generalization remains challenging:** While DSC-MAML improves cross-domain macro-F_1 over CNN-MAML, the absolute performance still lags the within-dataset setting. This suggests the need for more explicit domain adaptation mechanisms, such as domain-adversarial training or cross-attention fusion as in [14].

Current limitations include the reliance on flow-based features, which may miss fine-grained payload patterns, and the assumption of fixed feature spaces across datasets. Extending the framework to raw packet sequences and dynamically learned feature spaces is an important direction for future work.

5.4.1 Deployment considerations

Several limitations are central to the practical deployment of DSC-MAML in real NIDS environments. First, the system depends on a fixed, pre-defined feature space derived from flow statistics; any change in the flow exporter or feature extraction tool requires re-projection or retraining. Second, the assumption of flow-complete data introduces classification latency proportional to the flow duration (potentially seconds for long-lived flows), which may limit applicability to early-detection scenarios. Third, inner-loop gradient computation during inference-time adaptation incurs computational overhead that may preclude truly real-time use at line rate on high-speed links (10 Gbps and above) without hardware acceleration. Fourth, our evaluation is conducted on benchmark datasets and does not include live traffic evaluation; operational performance may differ due to concept drift, zero-day attacks, and traffic engineering effects not captured in static benchmarks. Future work will address these limitations through payload-aware extensions, streaming adaptation, and deployment on FPGA-based or Jetson-class edge hardware.

6 Conclusion and future work

This paper presented a meta-learning approach for few-shot network intrusion detection using a DSC backbone. By integrating a MAML objective with a parameter-efficient 1-D DSC architecture, the proposed DSC-MAML model learns an initialization that can rapidly adapt to attack types with only a handful of labeled examples.

Experiments on CICIDS2017, NSL-KDD and UNSW-NB15 demonstrated that DSC-MAML achieves higher accuracy and macro-F_1 than strong CNN-based and

metric-based baselines, while reducing the number of parameters by up to 65%. Future work will explore several directions. First, we plan to incorporate self-attention and cross-domain feature fusion modules into the DSC backbone, building on recent advances in attention-based few-shot NIDS [8, 11]. Second, we aim to investigate semi-supervised and self-supervised meta-learning strategies that can leverage large quantities of unlabeled traffic. Third, we will study robustness against adversarial examples and poisoning attacks in the meta-learning pipeline, as security-sensitive deployment requires resilience to adaptive adversaries. Finally, integrating the proposed framework into real-time NIDS and evaluating its performance on live traffic traces will be crucial to validate its practical utility.

Author contributions
H.A.-O.: Conceptualization, Methodology, Formal analysis, Writing - original draft. M.O.: Methodology, Validation, Writing - review & editing. A.A.: Data curation, Software, Writing - review & editing. J.S.A.: Investigation, Validation, Writing - review & editing. Y.A.: Software, Visualization, Writing - review & editing. S.A.: Data curation, Investigation, Writing - review & editing. N.A.: Formal analysis, Validation, Writing - review & editing. Q.A.-N.: Supervision, Conceptualization, Writing - review & editing. All authors reviewed and approved the final manuscript.

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Data availability
The datasets used in this study are publicly available at the following links: CICIDS2017 (https://www.unb.ca/cic/datasets/ids-2017.html); NSL-KDD (https://doi.org/10.5281/zenodo.17424143); and UNSW-NB15 (https://research.unsw.edu.au/projects/unsw-nb15-dataset).

Declarations

Ethical approval
Not applicable.

Consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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